Exercises (Systems, Determinants and Matrices)

1. Find all real x, y and z such that

$$\begin{cases} (x+1)yz = 27 \\ (y+1)zx = 12 \\ (z+1)xy = 12. \end{cases}$$

2. Let (x_1, y_1), (x_2, y_2) and (x_3, y_3) be three different real solutions to the system of equations $\begin{cases} x^3 - 5xy^2 = 21 \\ y^3 - 5x^2y = 28 \end{cases}$. Find the value of $\left(11 - \frac{x_1}{y_1}\right)\left(11 - \frac{x_2}{y_2}\right)\left(11 - \frac{x_3}{y_3}\right)$.

3. Solve in real numbers the system:

$$\begin{cases} x + y + z = a \\ x^2 + y^2 + z^2 = b^2 \\ xy - z^2 = 0. \end{cases}$$

Hence a and b are certain real numbers. Find conditions that the system has positive distinct solutions.

4. Find all real solutions of the system:

$$\begin{cases} x_5 + x_2 = yx_1 \\ x_1 + x_3 = yx_2 \\ x_2 + x_4 = yx_3 \\ x_3 + x_5 = yx_4 \\ x_4 + x_1 = yx_5, \end{cases}$$

where y is a real parameter.

5. Given the system with real coefficients:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0, \end{cases}$$

and such that (i) all a_{11}, a_{22}, a_{33} are positive; (ii) all other coefficients are negative; and (iii) for each row (equation), the sum of coefficients are positive.
Show that the system has only the trivial solution.

6. Find all real numbers x_1, x_2, x_3, x_4 for which the sum between each number and the product of the remaining numbers is 2.

7. Solve the system:

$$\begin{cases} |a_1 - a_2| x_2 + |a_1 - a_3| x_3 + |a_1 - a_4| x_4 = 1 \\ |a_2 - a_1| x_1 + |a_2 - a_3| x_3 + |a_2 - a_4| x_4 = 1 \\ |a_3 - a_1| x_1 + |a_3 - a_2| x_2 + |a_3 - a_4| x_4 = 1 \\ |a_4 - a_1| x_1 + |a_4 - a_2| x_2 + |a_4 - a_3| x_3 = 1, \end{cases}$$

Where a_1, a_2, a_3, a_4 are distinct real numbers.

8. Find all nonnegative solutions $(x_1, x_2, x_3, x_4, x_5)$ of the system of inequalities:

$$\begin{cases} (x_1^2 - x_3 x_5)(x_2^2 - x_3 x_5) \le 0 \\ (x_2^2 - x_4 x_1)(x_3^2 - x_4 x_1) \le 0 \\ (x_3^2 - x_5 x_2)(x_4^2 - x_5 x_2) \le 0 \\ (x_4^2 - x_1 x_3)(x_5^2 - x_1 x_3) \le 0 \\ (x_5^2 - x_2 x_4)(x_1^2 - x_2 x_4) \le 0. \end{cases}$$

9. Let

$$\begin{cases} a_{11} x_1 + a_{12} x_2 + \ldots + a_{1q} x_q = 0 \\ a_{21} x_1 + a_{22} x_2 + \ldots + a_{2q} x_q = 0 \\ \ldots \\ a_{p1} x_1 + a_{p2} x_2 + \ldots + a_{pq} x_q = 0 \end{cases}$$

be a system with p equations and q unknowns, with $q = 2p$, and $a_{ij} = \{-1, 0, 1\}$.

Show that there exists an *non-trivial integer* solution $(x_1, x_2, \ldots, x_q)$ such that $|x_j| \le q$ for all j.

10. Let $f(x)$ be a polynomial of degree 5 such that $f(1)=0$, $f(3)=1$, $f(9)=2$, $f(27)=3$, $f(81)=4$ and $f(243)=5$. Find the coefficient of x in $f(x)$.

11. The positive integers a and b are such that both $15a+16b$ and $16a-15b$ are squares. Find the least possible value that can be taken as the minimum of these two squares.

12. An $n\times n$ matrix whose entries come from the set $S=\{1,2,...,2n-1\}$ is a *silver* matrix if for each $i=1,2,...,n$, the i^{th} row and the i^{th} column together contains all elements of *S*. Show that
(a) there is no silver matrix for $n=1997$;
(b) silver matrices exist for infinitely many values of *n*.

13. Let there be *n* lamps $L_0, L_1, ..., L_{n-1}$ arranged in a circle. Each lamp is either ON or OFF. A sequence of steps $S_0, S_1, S_2, ..., S_i, ...$ is carried out. Step S_j affect the state of L_j only (leaving all other lamps unchanged) as follows:

(i) if L_{j-1} is ON, S_j changes the state of L_j;

(ii) if L_{j-1} is OFF, S_j leaves L_j unchanged.

The lamps are labelled mod *n*, i.e., $L_{-1}=L_{n-1}, L_0=L_n, L_1=L_{n+1}$, and so on. Initially all lamps are ON. Show that
(a) there is a positive integer $M(n)$ such that after $M(n)$ steps all the lamps are ON again.
(b) if $n=2^k$, then all the lamps are ON after n^2-1 steps.
(c) if $n=2^k+1$, then all the lamps are ON after n^2-n+1 steps.

14. In a chess tournament there are 100 players. On each day of the tournament, each player is designated to be ‘white’, ‘black’ or ‘idle’, and each white player will play a game with every black player. (You may assume that all games fixed for the day can be finished within that day.) At the end of the tournament, it was found day any two player have met exactly once. What is the minimum duration of the days that the tournament last?

15. In determinant Tic-Tac-Toe, player 1 enters 1 in an empty 3×3 matrix. Player 0 enters 0 a 0 in a vacant position, they play in turn until all entries are filled with five

1's and 4 0's. Player 0 wins if the determinant is 0, and player 1 wins otherwise. Assume both players play optimally, who has the winning strategy?

16. Let p be a prime. Show that the determinant of the matrix

$$\begin{pmatrix} x & y & z \\ x^p & y^p & z^p \\ x^{p^2} & y^{p^2} & z^{p^2} \end{pmatrix}$$

is congruent modulo p to a product of polynomials of the form $ax+by+cz$, where a, b and c are integers, namely the corresponding coefficients are congruent modulo p.

17. Define a sequence $\{u_n\}$ by $u_0 = u_1 = u_2 = 1$, and thereafter by the condition

$$\det\begin{pmatrix} u_n & u_{n+1} \\ u_{n+2} & u_{n+3} \end{pmatrix} = n!$$

for all $n \geq 0$. Show that u_n is an integer for all n. $(0! = 1.)$

18. Let H be an $n \times n$ matrix whose entries are ± 1 and whose rows are mutually orthogonal. Suppose H has an $a \times b$ submatrix whose entries are all 1. Show that $ab \leq n$.

19. Let d_n be the determinant of the $n \times n$ matrix whose entries from left to right, from top to bottom, are $\cos 1, \cos 2, \ldots, \cos n^2$. For example,

$$d_3 = \begin{vmatrix} \cos 1 & \cos 2 & \cos 3 \\ \cos 4 & \cos 5 & \cos 6 \\ \cos 7 & \cos 8 & \cos 9 \end{vmatrix}.$$

Find $\lim\limits_{n\to\infty} d_n$.

Note the angle measures are radian measures.

20. Let S be the set of all 2×2 real matrices

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

whose numbers a, b, c, d (in that order) form an arithmetic progression. Find all matrices M such that M in S for which there is some integer $k > 1$ such that M^k is also in S.

July2023

1. Find all real x, y and z such that

$$\begin{cases}(x+1)yz=27\\(y+1)zx=12\\(z+1)xy=12.\end{cases}$$

1. (August 2011 Test) Subtracting the first equation from the second, get $z(x-y)=-15$, second from the third, get $x(y-z)=0$ and first equation from the third, get $y(x-z)=-15$. The equation $x(y-z)=0$ implies $x=0$ or $y=z$. But $x=0$ will contradict $(y+1)zx=12$, hence we must have $y=z$. Putting $y=z$ into the original equations, we get $x=\dfrac{12}{y^2+y}$ and $x+1=\dfrac{27}{y^2}$. Eliminating x, we get $\dfrac{12+y^2+y}{y^2+y}=\dfrac{27}{y^2}$. As $y\neq 0$, we get $y^3+y^2-15y-27=0$. But $y^3+y^2-15y-27=(y+3)(y^2-2y+9)$, hence $y=-3$ or $y=1\pm\sqrt{10}$. From $y=-3$, get $x=2$. From $y=1\pm\sqrt{10}$, get $x=\dfrac{10}{3}-\dfrac{2}{3}(1\pm\sqrt{10})$. One checks these are indeed roots of the system. Hence the solutions of the system are $(x,y,z)=(2,-3,-3)$ or $(\frac{10}{3}-\frac{2}{3}\rho,\rho,\rho)$, where ρ is a root of $y^2-2y-9=0$.

2. Let (x_1,y_1), (x_2,y_2) and (x_3,y_3) be three different real solutions to the system of equations $\begin{cases}x^3-5xy^2=21\\y^3-5x^2y=28\end{cases}$. Find the value of $\left(11-\dfrac{x_1}{y_1}\right)\left(11-\dfrac{x_2}{y_2}\right)\left(11-\dfrac{x_3}{y_3}\right)$.

2. Observe that $4\left(x^3-5xy^2\right)=3\left(y^3-5x^2y\right)$. Divide both sides by y^3 and let $t=\dfrac{x}{y}$. Then $4\left(t^3-5t\right)-3\left(1-5t^2\right)=0$ which has three real roots (one between -5 and -4, another between -1 and 0, and the last one between 1 and 2. These roots are precisely $\dfrac{x_1}{y_1}$, $\dfrac{x_2}{y_2}$ and $\dfrac{x_3}{y_3}$ in some order.

Hence $4\left(t^3-5t\right)-3\left(1-5t^2\right)=4\left(t-\dfrac{x_1}{y_1}\right)\left(t-\dfrac{x_2}{y_2}\right)\left(t-\dfrac{x_3}{y_3}\right)$.

Therefore

$$\left(11 - \frac{x_1}{y_1}\right)\left(11 - \frac{x_2}{y_2}\right)\left(11 - \frac{x_3}{y_3}\right) = \frac{4\left(11^3 - 5\cdot 11\right) - 3\left(1 - 5\cdot 11^2\right)}{4} = 1729$$

3. Solve in real numbers the system:

$$\begin{cases} x+y+z=a \\ x^2+y^2+z^2=b^2 \\ xy-z^2=0. \end{cases}$$

Hence a and b are certain real numbers. Find conditions that the system has positive distinct solutions.

3. (IMO 1961) Assume $a \neq 0$. Then $z = \dfrac{a^2-b^2}{2a}$. Use this to get $x+y = \dfrac{a^2+b^2}{2a}$, $xy = \dfrac{(a^2-b^2)}{4a^2}$. Finally the solutions can be obtained. These are two solutions and are positive if and only if $a>0$ and $|a|>|b|$. The solutions exist if and only if the sphere $x^2+y^2+z^2=b^2$ and the plane $x+y+z=a$ intersect. Then the intersection circle will intersect the hyperbolic surface $xy=z^2$, since the surface contains the origin and is unbounded. From this get $\dfrac{|a|}{\sqrt{3}}<|b|$. Combined to get $|b|<a<|b|/\sqrt{3}$.

4. Find all real solutions of the system:

$$\begin{cases} x_5+x_2=yx_1 \\ x_1+x_3=yx_2 \\ x_2+x_4=yx_3 \\ x_3+x_5=yx_4 \\ x_4+x_1=yx_5, \end{cases}$$

where y is a real parameter.

4. (IMO1963) Write this as a system of equations of x, we get the determinant

$$\begin{vmatrix} -y & 1 & 0 & 0 & 1 \\ 1 & -y & 1 & 0 & 0 \\ 0 & 1 & -y & 1 & 0 \\ 0 & 0 & 1 & -y & 1 \\ 1 & 0 & 0 & 1 & -y \end{vmatrix} = (2-y)(y^2+y-1)^2.$$

It has unique solution if $y \neq 2$, and $y^2+y-1 \neq 0$. Otherwise infinitely many solutions.

5. Given the system with real coefficients:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0, \end{cases}$$

and such that (i) all a_{11}, a_{22}, a_{33} are positive; (ii) all other coefficients are negative; and (iii) for each row (equation), the sum of coefficients are positive.
Show that the system has only the trivial solution.

5. (IMO 1965) Suppose the system has non-trivial solution (x_1, x_2, x_3) and suppose $|x_1| \geq |x_2| \geq |x_3|$, we must have $|x_1| > 0$. By dividing, get

$$a_{11} + a_{12}\frac{x_2}{x_1} + a_{13}\frac{x_3}{x_1} \geq a_{11} + a_{12}\left|\frac{x_2}{x_1}\right| + a_{13}\left|\frac{x_3}{x_1}\right| \geq a_{11} - |a_{12}| - |a_{13}| = a_{11} + a_{12} + a_{13} > 0.$$

Since (x_1, x_2, x_3) is a solution,

$0 = |a_{11}x_1 + a_{12}x_2 + a_{13}x_3| = |x_1|\left|a_{11} + a_{12}\frac{x_2}{x_1} + a_{13}\frac{x_3}{x_1}\right| > 0$, a contradiction. Thus $x_1 = 0$, and hence $x_2 = x_3 = 0$.

6. Find all real numbers x_1, x_2, x_3, x_4 for which the sum between each number and the product of the remaining numbers is 2.

6. (IMO1965) We have

$$\begin{cases} x_1 + x_2x_3x_4 = 2 \\ x_2 + x_1x_3x_4 = 2 \\ x_3 + x_1x_2x_4 = 2 \\ x_4 + x_1x_2x_3 = 2. \end{cases}$$

Note $x_i \neq 0$ for all i. Let $x_1x_2x_3x_4 = p$. Then the system becomes $x_i + \frac{p}{x_i} = 2$, for all i. Hence all x_i are solutions of $x^2 - 2x + p = 0$. Write this as $(x-1)^2 = 1 - p$.

For $p = 1$, all x_i are 1. For $p < 1$, we have two cases. (a) $x_1 = x_2, x_3 = x_4$ are two distinct pairs of roots. The system becomes $x_1 + x_1x_3^2 = 2$ and $x_3 + x_1x_3^2 = 2$. By subtracting, get $(x_1 - x_3)(1 - x_1x_3) = 0$. Because $x_1x_3 = \sqrt{p} < 1$, get $x_1 = x_3$, a contradiction. (b) $x_1 = x_2 = x_3$ and $x_4 \neq x_1$. For $x_1 = x_2 = x_3 = 1$, get $x_4 = 1$. For $x_1 = x_2 = x_3 = -1$, get $x_4 = 3$.

7. Solve the system:

$$\begin{cases} |a_1 - a_2|x_2 + |a_1 - a_3|x_3 + |a_1 - a_4|x_4 = 1 \\ |a_2 - a_1|x_1 + |a_2 - a_3|x_3 + |a_2 - a_4|x_4 = 1 \\ |a_3 - a_1|x_1 + |a_3 - a_2|x_2 + |a_3 - a_4|x_4 = 1 \\ |a_4 - a_1|x_1 + |a_4 - a_2|x_2 + |a_4 - a_3|x_3 = 1, \end{cases}$$

Where a_1, a_2, a_3, a_4 are distinct real numbers.

7. (IMO1966). The system is symmetric, so assume $a_1 > a_2 > a_3 > a_4$, get

$$\begin{cases} (a_1 - a_2)x_2 + (a_1 - a_3)x_3 + (a_1 - a_4)x_4 = 1 \\ (a_1 - a_2)x_1 + (a_2 - a_3)x_3 + (a_2 - a_4)x_4 = 1 \\ (a_1 - a_3)x_1 + (a_2 - a_3)x_2 + (a_3 - a_4)x_4 = 1 \\ (a_1 - a_4)x_1 + (a_2 - a_4)x_2 + (a_3 - a_4)x_3 = 1. \end{cases}$$

By subtracting each equation from the previous one (except the last equation untouched), we get

$$\begin{cases} (a_1-a_2)(-x_1+x_2+x_3+x_4)=0 \\ (a_2-a_3)(-x_1-x_2+x_3+x_4)=0 \\ (a_3-a_4)(-x_1-x_2-x_3+x_4)=0 \\ (a_1-a_4)x_1+(a_2-a_4)x_3+(a_3-a_4)x_3=1. \end{cases}$$

It has the unique solution $x_1 = x_4 = \dfrac{1}{a_1-a_4}, x_2 = x_3 = 0.$

8. Find all nonnegative solutions $(x_1, x_2, x_3, x_4, x_5)$ of the system of inequalities:

$$\begin{cases} (x_1^2 - x_3x_5)(x_2^2 - x_3x_5) \le 0 \\ (x_2^2 - x_4x_1)(x_3^2 - x_4x_1) \le 0 \\ (x_3^2 - x_5x_2)(x_4^2 - x_5x_2) \le 0 \\ (x_4^2 - x_1x_3)(x_5^2 - x_1x_3) \le 0 \\ (x_5^2 - x_2x_4)(x_1^2 - x_2x_4) \le 0. \end{cases}$$

8. (IMO 1972). The first inequality can be written as $(x_1x_2)^2 + (x_3x_5)^2 - x_3x_5(x_1^2 + x_2^2) \le 0.$ Others are by circular permutations. Adding the five inequalities we get:

$$\frac{1}{2}\sum x_1^2((x_2 - x_4)^2 + (x_3 - x_5)^2) \le 0,$$

the sum extended to all circular permutations. Thus we obtain $x_2 = x_4, x_3 = x_5, x_4 = x_1, x_5 = x_2,$ until all x_i are equal to a certain real number α.

9. Let

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1q}x_q = 0 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2q}x_q = 0 \\ \ldots \\ a_{p1}x_1 + a_{p2}x_2 + \ldots + a_{pq}x_q = 0 \end{cases}$$

be a system with p equations and q unknowns, with $q = 2p$, and $a_{ij} = \{-1, 0, 1\}$.

Show that there exists an *non-trivial integer* solution $(x_1, x_2, \ldots, x_q)$ such that $|x_j| \le q$

for all j.

9. (IMO1976). Let $A=\{x\in\mathbb{Z}\,\big|\,|x|\leq p\}$ and $B=\{y\in\mathbb{Z}\,\big|\,|y|\leq 2p^2\}$ be two set of integers. We have $|A|=2p+1$ and $|B|=4p^2+1$. Let $f:\mathbb{Z}^{2p}\to\mathbb{Z}^{p}$ be the linear map given by the matrix (a_{ij}) such that $f(x_1,x_2,...,x_q)=(\sum_{j=1}^{q}a_{1j}x_j,...,\sum_{j=1}^{q}a_{pj}x_j).$ Clearly $f(\mathbb{Z}^{2p})\subseteq\mathbb{Z}^{p}$. Indeed $f(A^{2p})\subseteq B^p$ because

$$\left|a_{i1}x_1+a_{i2}x_2+...+a_{iq}x_q\right|\left|a_{i1}\right|\left|x_1\right|+\left|a_{i2}\right|\left|x_2\right|+...+\left|a_{iq}\right|\left|x_q\right|\leq\left|x_1\right|+\left|x_2\right|+...+\left|x_q\right|\leq pq=2p^2.$$

Now it suffices to show $\left|B^p\right|<\left|A^{2p}\right|$ then we can find two distinct elements $(x_1,x_2,...,x_q)$ and $(x_1',x_2',...,x_q')$ then have the same image, and hence $f(x-x')=0$ and with $x-x'$ a solution required, as $\left|x_i-x_i'\right|\leq\left|x_i\right|+\left|x_i'\right|\leq 2p=q$. The claimed inequality is easy to check as $(4p^2+1)^p\leq(2p+1)^{2p}$.

10. Let $f(x)$ be a polynomial of degree 5 such that $f(1)=0,\quad f(3)=1,\quad f(9)=2,$ $f(27)=3,\quad f(81)=4$ and $f(243)=5.$ Find the coefficient of x in $f(x).$

Define $g(x)=f(3x)-f(x)-1$. Then $g(1)=\mathrm{g}(3)=\mathrm{g}(9)=\mathrm{g}(27)=\mathrm{g}(81)=0$. As $g(x)$ is a polynomial with degree at most 5, we have $g(x)=k(x-1)(x-3)(x-9)(x-27)(x-81)$ for some constant k. Since $g(0)=-1$, we have $k=\dfrac{1}{1\times3\times9\times27\times81}$. In other words, we have

$g(x)=\dfrac{(x-1)(x-3)(x-9)(x-27)(x-81)}{1\times3\times9\times27\times81}$ and the coefficient of x is

$\dfrac{1}{81}+\dfrac{1}{27}+\dfrac{1}{9}+\dfrac{1}{3}+\dfrac{1}{1}$. Note that if the coefficient of x in $f(x)$ is c, then the same coefficient in $f(3x)$ would be $3c$ and hence that in $g(x)$ would be $2c$. In other words, the coefficient of x in $f(x)$ is half of that in $g(x)$. Therefore the answer is

$$\frac{1}{2}\left(\frac{1}{81}+\frac{1}{27}+\frac{1}{9}+\frac{1}{3}+\frac{1}{1}\right)=\frac{121}{162}.$$

Remark. Clearly a solution using brute force exists but it is extremely tedious. Indeed it can be handled using sophisticated software.

11. The positive integers a and b are such that both $15a+16b$ and $16a-15b$ are squares. Find the least possible value that can be taken as the minimum of these two squares.

11. (IMO1996). Solving the system:

$$\begin{cases} 15a+16b=x^2 \\ 16a-15b=y^2, \end{cases}$$

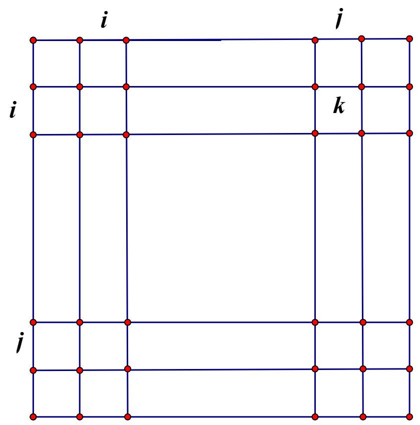
get $a=\dfrac{15x^2+16y^2}{481}; b=\dfrac{16x^2-15y^2}{481}.$ From $481=13\times 37,$ we consider the system

$$\begin{cases} 15x^2+16y^2 \equiv 0 \pmod{13} \\ 15x^2+16y^2 \equiv 0 \pmod{37}. \end{cases}$$

Using the law of quadratic reciprocity, or otherwise, we show $x \equiv y \equiv 0 \pmod{13}$ and $x \equiv y \equiv 0 \pmod{37}$. Hence both x and y are multiples of 481. When $x=y=481,$ get the smallest solution $a=31\times 481$ and $b=481.$ The answer is $481^2.$

12. An $n\times n$ matrix whose entries come from the set $S=\{1,2,...,2n-1\}$ is a *silver* matrix if for each $i=1,2,...,n,$ the i^{th} row and the i^{th} column together contains all elements of S. Show that
(a) there is no silver matrix for $n=1997$;
(b) silver matrices exist for infinitely many values of n.

12. (IMO1997). (a) We call a *cross* C_i the union of the i^{th} row and the i^{th} column of A. Thus A has n crosses, all of them contains the elements $1,2,...,2n-1.$ As the diagonal of A contains at most n elements, there exists k**,** which does not appear in the diagonal. Note every cross contains the *unique* element k. In the cross C_i, say k appears as a_{ij}, namely in the $i\times j$ position, with $i\neq j.$ Thus k also appear in column j, namely in the cross C_j. Therefore C_i, C_j pair up to form an equivalence class. Using the same k, all crosses are paired up, then n is even.



(b) We show that silver matrices exist for $2^n \times 2^n$ matrices. For $n=1$, we have the matrix: $\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$. For $n=2$, we have

$$\begin{pmatrix} 1 & 2 & 4 & 5 \\ 3 & 1 & 5 & 4 \\ 6 & 7 & 1 & 2 \\ 7 & 6 & 3 & 1 \end{pmatrix}.$$

Note how it is constructed from the old. For $n=3$, we have

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 8 & 9 & 10 & 11 \\ 3 & 1 & 5 & 4 & 9 & 10 & 11 & 8 \\ 6 & 7 & 1 & 2 & 10 & 11 & 8 & 9 \\ 7 & 6 & 3 & 1 & 11 & 8 & 9 & 10 \\ 12 & 13 & 14 & 15 & 1 & 2 & 4 & 5 \\ 13 & 14 & 15 & 12 & 3 & 1 & 5 & 4 \\ 14 & 15 & 12 & 13 & 6 & 7 & 1 & 2 \\ 15 & 12 & 13 & 14 & 7 & 6 & 3 & 1 \end{pmatrix}.$$

While the $2^n \times 2^n$ matrix A_n (consisting of elements from 1 to $2^{n+1}-1$) is constructed, the $2^{n+1} \times 2^{n+1}$ matrix A_{n+1} is constructed as $\begin{pmatrix} A_n & B_n \\ C_n & A_n \end{pmatrix}$, where A_n is as the old, and B_n is the $2^n \times 2^n$ matrix consisting of elements from 2^{n+1} to $2^{n+1}+2^n-1$, as

$$\begin{pmatrix} 2^{n+1} & 2^{n+1}+1 & . \ . & 2^{n+1}+2^n-1 \\ 2^{n+1}+1 & 2^{n+1}+2 & . \ . & 2^{n+1} \\ . & . & . \ . & . \\ . & 2^{n+1}+2^n-1 & . \ . & . \\ 2^{n+1}+2^n-1 & 2^{n+1} & . \ . & 2^{n+1}+2^n-2 \end{pmatrix}.$$

Also C_n is the $2^n \times 2^n$ matrix consisting of elements from $2^{n+1}+2^n$ to $2^{n+2}-1$, as

$$\begin{pmatrix} 2^{n+1}+2^n & 2^{n+1}+2^n+1 & . \ . & 2^{n+2}-1 \\ 2^{n+1}+2^n+1 & 2^{n+1}+2^n+2 & . \ . & 2^{n+1}+2^n \\ . & . & . \ . & . \\ . & 2^{n+2}-1 & . \ . & . \\ 2^{n+2}-1 & 2^{n+1}+2^n & . \ . & 2^{n+2}-2 \end{pmatrix}.$$

13. Let there be *n* lamps $L_0, L_1, ..., L_{n-1}$ arranged in a circle. Each lamp is either ON or OFF. A sequence of steps $S_0, S_1, S_2, ..., S_i, ...$ is carried out. Step S_j affect the state of L_j only (leaving all other lamps unchanged) as follows:

(i) if L_{j-1} is ON, S_j changes the state of L_j;

(ii) if L_{j-1} is OFF, S_j leaves L_j unchanged.

The lamps are labelled mod *n*, i.e., $L_{-1} = L_{n-1}, L_0 = L_n, L_1 = L_{n+1}$, and so on. Initially all lamps are ON. Show that

(a) there is a positive integer $M(n)$ such that after $M(n)$ steps all the lamps are ON again.

(b) if $n = 2^k$, then all the lamps are ON after $n^2 - 1$ steps.

(c) if $n = 2^k + 1$, then all the lamps are ON after $n^2 - n + 1$ steps.

13. (IMO 1993). The state of the system of the lamps may be described by the vector $v \in F_2^n, v = (x_0, x_2, ..., x_{n-1})$, where $x_j = 1$ if L_j is on, and $x_j = 0$ if L_j is off. The step S_j is like $v = (x_0, x_1, ..., x_{n-1}) \mapsto S_j(v) = (x_0, x_1, ... x_{j-1}, x_{j-1} + x_j, ..., x_{n-1})$. In terms

of matrices S_j is the operator from $F_2^n \to F_2^n$:

$$\begin{pmatrix} 1 & 0 & \dots & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 1 & & & \\ & & 1 & 1 & & \\ & & & & & \\ & & & & & 1 \end{pmatrix},$$

where the two "1" occurs at the j^{th} row of course. Let $P : F_2^n \to F_2^n$ is the *rotation* operator $P(x_0, x_1, ..., x_{n-1}) = (x_2, x_3, ..., x_n, x_0)$, or the matrix:

$$\begin{pmatrix} 0 & 1 & \dots & \dots & \dots & 0 \\ 0 & 0 & 1 & \dots & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & 0 \\ & & \dots & \dots & \dots & \dots \\ & & & & 0 & 1 \\ 1 & & & & & 0 \end{pmatrix}.$$

Looking carefully at the coordinates of the vector v it is easy to prove that $S_j = P^{-j} S_0 P^j$. Thus after the steps $S_0, S_1, ..., S_{j-1}$ are carried out, the initial vector v are carried out, the initial vector v is mapped as follows:

$$S_{j-1} S_{j-2} ... S_0(v) = (P^{-(j-1)} S_0 P^{j-1})(P^{-(j-2)} S_0 P^{j-2})...(P^{-1} S_0 P) S_0(v) = P^{-j} (PS_0)^j (v).$$

The composed vector $(PS_0)(v) = P(x_0 + x_{n-1}, x_1, ..., x_{n-1}) = (x_1, ..., x_{n-1}, x_0 + x_{n-1})$. Thus it is represented as

$$PS_0 = \begin{pmatrix} 0 & 1 & 0 & . & . & 0 \\ 0 & 0 & 1 & . & . & 0 \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ 0 & 0 & 0 & . & . & 1 \\ 1 & 0 & 0 & . & . & 1 \end{pmatrix} = A.$$

Using the *Cayley-Hamilton* theorem, A satisfies the equation $F(X) = \det(A - XI)$.

Direct computations show

$$F(X)=\begin{vmatrix} -X & 1 & 0 & . & . & 0 \\ 0 & -X & 1 & 0 & . & 0 \\ & \dots & \dots & \dots & \dots & \dots \\ & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & -X & 1 \\ 1 & 0 & 0 & \dots & \dots & 1-X \end{vmatrix} = (-1)^n(X^n - X^{n-1} - 1).$$

(Expand by the first column.)

As we are working in F_2, $A^n = A^{n-1} + I$.

(a) As the operators are elements of finite group, we can find j such that $P^{-j}(PS_0)^j(v) = v$.

(b) For $n = 2^k$. Then $A^{n^2} = (A^{n-1} + I)^n = A^{n(n-1)} + I$. So

$I = A^{n^2} + A^{n(n-1)} = A^{n^2-n}(A^n + I) = A^{n^2-n}A^{n-1} = A^{n^2-1}$.

(c) Let $n = 2^k + 1$. Then

$A^{n^2-1} = (A^{n+1})^{n-1} = (A^n + A)^{n-1} = A^{n(n-1)} + A^{n-1}$.

Hence $A^{n-1} = A^{n^2-1} + A^{n^2-n} = A^{n^2-n}(A^{n-1} + I) = A^{n^2-n}A^n = A^{n^2}$. Hence $A^{n^2-n+1} = I$ as required.

(Note uses of the binomial theorem often).

14. In a chess tournament there are 100 players. On each day of the tournament, each player is designated to be 'white', 'black' or 'idle', and each white player will play a game with every black player. (You may assume that all games fixed for the day can be finished within that day.) At the end of the tournament, it was found day any two player have met exactly once. What is the minimum duration of the days that the tournament last?

14. This is actually the theorem of Graham-Pollack, a proof may be found in the book: A course in Combinatorics, Van Lint and Wilson.

Assign variable x_i to player $i, 1 \le i \le 100$. Clearly 99 days suffice. On the first day player 1 (white) against players 2, 3, …, 100 (black). On the second day player 2 (white) against player 3, 4, …, 100, and so on, until on the 99th day, player 99 (white) against player 100 (black). So 99 days suffice.

Suppose now the game may be player in (strictly) fewer than 99 days. We form the system of equations. The first equation is the sum of all those variables who correspond the white players played on the first day, and we assume they sum to 0. Likewise we do it for the second day and so on, and we get 98 or fewer equations. Finally we sum all those variables $x_1 + x_2 + \ldots + x_{100} = 0$. So we get a system of 99 or fewer homogeneous linear equations. Because we have 100 variables, thus we have a non-trivial (nonzero) solution $a_1, a_2, \ldots, a_{100}$. Hence

$$0 = (a_1 + a_2 + \ldots + a_{100})^2 = (a_1^2 + a_2^2 + \ldots + a_{100}^2) + 2 \sum_{1 \le i < j \le 100} a_i a_j .$$

Since any two players have met exactly once, we have

$$\sum_{1 \le i < j \le 100} a_i a_j = \sum_k \Big(\sum_{w \in W_k} a_w \sum_{b \in B_k} b_k \Big),$$

where W_k and B_k are the set of white and black players on day k respectively. But we have $\sum_{w \in W_k} a_w = 0$ on every day k, hence we have

$0 = (a_1 + a_2 + \ldots + a_{100})^2 = (a_1^2 + a_2^2 + \ldots + a_{100}^2) \ne 0$, a contradiction.

15. In determinant Tic-Tac-Toe, player 1 enters 1 in an empty 3×3 matrix. Player 0 enters 0 a 0 in a vacant position, they play in turn until all entries are filled with five 1’s and 4 0’s. Player 0 wins if the determinant is 0, and player 1 wins otherwise. Assume both players play optimally, who has the winning strategy?

(Putnam 2002, player 1 cannot prevent player 0 getting a row of 0’s, or a column of 0’s , or a 2×2 submatrix of 0s.

16. Let p be a prime. Show that the determinant of the matrix

$$\begin{pmatrix} x & y & z \\ x^p & y^p & z^p \\ x^{p^2} & y^{p^2} & z^{p^2} \end{pmatrix}$$

is congruent modulo p to a product of polynomials of the form $ax+by+cz,$ where a, b and c are integers, namely the corresponding coefficients are congruent modulo p.

(Putnam 2002). The determinant is congruent modulo p to

$$x\prod_{i=0}^{p-1}(y+ix)\prod_{i,j=0}^{p-1}(z+ix+jy).$$

17. Define a sequence $\{u_n\}$ by $u_0=u_1=u_2=1,$ and thereafter by the condition

$$\det\begin{pmatrix} u_n & u_{n+1} \\ u_{n+2} & u_{n+3} \end{pmatrix}=n!$$

for all $n\geq 0.$ Show that u_n is an integer for all n. $(0!=1.)$

(Putnam 2004) Let $v_n=(n-1)(n-3)...(4)(2),$ if n odd, and $v_n=(n-1)(n-3)...(3)(1),$ if n even. Show that $u_n=v_n$ by induction.

18. Let H be an $n\times n$ matrix whose entries are ± 1 and whose rows are mutually orthogonal. Suppose H has an $a\times b$ submatrix whose entries are all 1. Show that $ab\leq n.$

(Putnam 2005) Choose a set of a rows $r, r_2, ..., r_a$ containing the $a\times b$ submatrix whose entries are 1. Then for $i,j=1,2,...,a,$ $r_i\cdot r_j=n$ if $i=j$ and $r_i\cdot r_j=0$ otherwise. Hence $\sum_{i,j=1}^{a} r_i\cdot r_j=an.$ On the other hand, the term of the left is the dot product of $r_1+r_2+...+r_a$ with itself, or its square length. Since this vector has a 1's in each of b of its coordinates, hence its product is at least $a^2b.$ Hence $an\geq a^2b,$ hence the result.

19. Let d_n be the determinant of the $n\times n$ matrix whose entries from left to right, from top to bottom, are $\cos 1, \cos 2, ..., \cos n^2.$ For example,

$$d_3=\begin{vmatrix} \cos 1 & \cos 2 & \cos 3 \\ \cos 4 & \cos 5 & \cos 6 \\ \cos 7 & \cos 8 & \cos 9 \end{vmatrix}.$$

Find $\lim_{n\to\infty} d_n.$

Note the angle measures are radian measures.

(Putnam 2009) Adding the third column to the first, show that $d_n = 0$ for $n \geq 3$.

20. Let *S* be the set of all 2×2 real matrices

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

whose numbers *a*, *b*, *c*, *d* (in that order) form an arithmetic progression. Find all matrices *M* such that *M* in *S* for which there is some integer $k > 1$ such that M^k is also in *S*.

(Putnam 2015) They are the scalar multiples (not linear combinations) of $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -3 & -1 \\ 1 & 3 \end{pmatrix}$. Any element of *S* may be written as $M = \alpha A + \beta B$, with $\alpha, \beta \in \mathbb{R}$. Note that $A^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$ and $B^3 = \begin{pmatrix} -24 & -8 \\ 8 & 24 \end{pmatrix}$ are both in *S*, so is its scalar multiples. Indeed these are the only cases. Suppose $M = \alpha A + \beta B$, with α, β nonzero. Let $C = \begin{pmatrix} 1 & 1/\sqrt{2} \\ -1 & 1/\sqrt{2} \end{pmatrix}$, then $C^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$. Let $D = C^{-1}MC$, then $D = 2\alpha \begin{pmatrix} 0 & \gamma \\ \gamma & 1 \end{pmatrix}$, where $\gamma = -\dfrac{\beta\sqrt{2}}{\alpha}$. Suppose now M^k is in *S* with $k \geq 2$. The equalities $\begin{pmatrix} 1 & -1 \end{pmatrix} A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \end{pmatrix} B \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$ implies $\begin{pmatrix} 1 & -1 \end{pmatrix} M \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$, and so the upper left entry of $C^{-1}M^kC = D^k$ is 0. On the other hand, from the expression for *D*, and by induction on *k*, we have $D^k = (2\alpha)^k \begin{pmatrix} \gamma^2 p_{k-1} & \gamma p_k \\ \gamma p_k & p_{k+1} \end{pmatrix}$, where p_k is defined recursively by $p_0 = 0,\ p_1 = 1,\ \ p_{k+2} = \gamma^2 p_k + p_{k+1}$. Hence $p_k > 0$ when $k \geq 1$, whence the upper left entry of D^k is nonzero, a contradiction.

July 2023